Raymond Dacosta Azadda

Biostatistician
Center for Alaska Native Health Research, Institute of Arctic Biology
University of Alaska Fairbanks

2141 Koyukuk Drive, Room 222A, Fairbanks, AK 99775

rdazadda.github.io

RESEARCH INTERESTS

Methodology

Multilevel and Longitudinal Analysis; Bayesian Methods; Machine Learning; Survey Design; Factor Analysis; Psychometrics; Q Methodology; Qualitative Research; Accelerometry/Actigraphy Analysis; Circadian Rhythm Analysis; Open-Source Statistical Software; Applied Statistics

Applied Research

American Indian/Alaska Native Research; Community-Based Participatory Research (CBPR); Heart Disease and Chronic Disease Prevention; Suicide Prevention; Sleep and Physical Activity Research; Behavioral Health; Substance Use; Environmental Health; Applied Psychometrics

POSITIONS

Biostatistician 08/2026 - present
Center for Alaska Native Health Research, University of Alaska Fairbanks Fairbanks, AK

Biostatistician (Temporary Employee) 07/2025 - 08/2026
Center for Alaska Native Health Research, University of Alaska Fairbanks Fairbanks, AK

Core Member, Community-Based Participatory Research (CBPR) Core 02/2026 - present
Alaska INBRE (IDeA Network of Biomedical Research Excellence) Fairbanks, AK

Graduate Teaching Assistant 08/2023 - 05/2025
University of Alaska Fairbanks Fairbanks, AK

Teaching and Research Assistant 11/2022 - 08/2023
University of Cape Coast Cape Coast, Ghana

EDUCATION

M.S., Statistics and Data Science 08/2023 - 05/2025
University of Alaska Fairbanks Fairbanks, AK

B.Sc., Actuarial Science 08/2018 - 11/2022
University of Cape Coast Cape Coast, Ghana

FUNDED RESEARCH

Center for Alaska Native Health Research (CANHR), University of Alaska Fairbanks

Current

1. **Qungasvik (Toolbox)-Healthy Hearts: An Alaska Enrichment Initiative (AK Enrichment)** 2025 - present
   PI: Dr. Stacy Rasmus · Funder: **NHLBI (OT2HL158287)** · Qungasvik website · AK CEAL project page · NIH CEAL Network

   Role Member of the project's Biostatistics ("Numbers") Team, contributing to survey design and quantitative analyses, including multilevel modeling, physical activity analysis, and Bayesian methods. Out of this project, developed the CANHRActi dashboard and desktop apps and the actiRhythm package.

   About A culturally grounded, evidence-based intervention promoting traditional Yup'ik diet, physical activity, and social connection for Alaska Native Elders and youth.

2. **Alaska Alliance for Community Engagement - Partnership for Action Toward Health (AK ACE - PATH)** 2025 - present
PI: Dr. Stacy Rasmus · Funder: **NHLBI (OT2HL158287)** · AK ACE - PATH project page

Role Member of the project's Biostatistics ("Numbers") Team, contributing to quantitative analyses, including multilevel modeling, Q-methodology, and Bayesian methods. Out of this project, developed the canhrQsort dashboard and the bayesqm package.

About A research initiative implementing localized solutions to promote health and prevent chronic disease in rural Alaska Native communities, prioritizing Indigenous knowledge.

3. **Neqpiaput Iinruugut (Our Food is Medicine)** 2026 - present
PI: Dr. Andrea Bersamin · Funder: **NHLBI (R01HL181633-01)**

Role One of the project's two statisticians, contributing to survey design and quantitative analyses, including multilevel modeling and other statistical methods.

About A tribally-driven Indigenous food sovereignty intervention to prevent coronary heart disease among Yup'ik Alaska Native communities in the Yukon-Kuskokwim Delta region.

4. **Native Transformations Project (NTP)** 2025 - present
PI: Dr. Stacy Rasmus · Funder: **NIH** · Intertribal Healthy Data Dashboard

Role Member of the project's Biostatistics ("Numbers") Team, contributing to the project's analysis and developing the Intertribal Healthy Data Dashboard, an online platform for displaying community level health information and community level protective factors.

About A Tribally initiated, directed, and controlled research study on strengths, protections, and resilience against substance abuse.

5. **Alaska INBRE Community-Based Participatory Research (CBPR) Core** 2026 - present
PI: Dr. Katie Cueva · Funder: **NIH**, through the Alaska INBRE (IDeA Network of Biomedical Research Excellence) · CBPR Toolkit

Role Member of the CBPR Core, conducting semi-structured interviews with CBPR researchers across Alaska, performing thematic analysis of interview transcripts, and helping produce one-pagers that synthesize community-engaged research practices into the Core's CBPR Toolkit.

About A core within the Alaska INBRE that promotes equitable, inclusive, and sustainable community-based participatory research across Alaska.

Completed

1. **Preventing the Disproportionate Incarceration and Recidivism of Alaska Native People** 2025 - 2026
PI: Dr. Katie Cueva · Funder: **Alaska Federation of Natives** · Final Report to the Alaska Legislature

Role Statistician on the UAF CANHR team, one of the study's partner teams, analyzing over 80 survey responses and contributing to thematic analysis of over 20 expert interview transcripts for the final stakeholder report to the Alaska Legislature.

About Investigated factors driving the disproportionate incarceration of Alaska Native people (40% of Alaska's prison population despite making up 14% of the state population).

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Hueffer, K., Brinkman, T., Robinson, H. D., Joseph, V., Chaliak, J. A., John, S., Clinton, M., Shaw, J., Peter, T., Bersamin, A., Allen, J., **Azadda, R. D.**, Grogan-Kaylor, A., Lee, K., & Rasmus, S. (2026). Health priorities for rural Alaska Native communities at the extremes of weather and climate change. *BMC Medicine*. doi

Manuscripts Under Review

1. **Azadda, R. D.**, Agbi-Kaiser, H. O., Robinson, H. D., Lee, K., AKE-ACE Team, Hueffer, K., Peter, T., & Rasmus, S. (2026). Quantifying Uncertainty in Q Methodology: Bayesian Factor Analysis from an Exact Rank-Order Likelihood. Submitted to *Behavior Research Methods*.
2. Clinton, M., **Azadda, R. D.**, Pender, J. M., Wing, A., Glass, J. R., & Muehlbauer, J. D. (2026). An eDNA-inspired, non-destructive approach to DNA isolation from archived fish specimen fixative media. Submitted to *Natural History Collections and Museomics*.
3. Grogan-Kaylor, A., Lee, K., Allen, J., Qungasvik Team, **Azadda, R. D.**, & Rasmus, S. (2026). New Perspectives on Alcohol's Relation to Suicide: Intervention Effects of Growth in Suicide Protection Produces Growth in Alcohol Protection in Rural Alaska Native Youth. Submitted to *Journal of the Society for Social Work and Research*.

REPORTS

1. Alaska Federation of Natives. (2025). Recommendations to Reduce the Overrepresentation of Alaska Native People in the Criminal Justice System. Final report to the Alaska Legislature.

PRESENTATIONS AND POSTERS

Presentations

1. Rowles, F., Cueva, K., Kloster, K., Thompson, L., **Azadda, R. D.**, Kiker, L., Chaliak, A. J., & Coble, A. (2026). Recommendations to Reduce the Overrepresentation of Alaska Native people in Alaska's Criminal Legal System. Oral presentation, APHA 2026 Annual Meeting and Expo. Abstract

Posters

1. Robinson, H. D., Hueffer, K., Peter, T., Brinkman, T., Clinton, M., Bersamin, A., Charles, W., Joseph, V., John, S., Chaliak, A. J., **Azadda, R. D.**, Allen, J., Lee, K., Grogan-Kaylor, A., Rowles, F., & Rasmus, S. (2026). An adapted version of Q-methodology provides an in-depth understanding of the impact of climate change and health on rural Alaskan communities. Poster, 2026 CEAL Annual Meeting. View poster
2. Rasmus, S., Cueva, K., Robinson, H. D., Grogan-Kaylor, A., O'Brien, D., Bersamin, A., Allen, J., Lee, K., **Azadda, R. D.**, Chaliak, A. J., Johnson, J., Joe, J., Rowles, F., & Shaw, J. (2026). The Qungasvik-Healthy Hearts Intervention: A culturally grounded and evidence-based intervention to support increased physical activity, a traditional Yup'ik diet, and social connection for Elders and youth in Alaska. Poster, 2026 CEAL Annual Meeting. View poster

RESEARCH SOFTWARE

Dashboards and Desktop Apps

1. **CANHRActi** Analysis of accelerometer data for physical activity, sleep, sedentary behavior and circadian rhythm research 2025 - present
   Open dashboard · Windows · macOS (Apple silicon) · macOS (Intel) · Linux · GitHub
   A web and desktop app that does the work of GGIR and of count-based accelerometer software in one place: its raw pipeline implements GGIR's methods, and it runs the wear-time, sleep-scoring, cut-point and energy-expenditure algorithms that count-based software uses on ActiGraph files. It then goes further, with advanced circadian rhythm and sedentary fragmentation analysis.
2. **canhrQsort** A dashboard for Q-sort analysis 2026
   Open dashboard · GitHub
   An online dashboard that ranks the statements of a Q-sort study by average placement, compares rankings across demographic groups, and identifies shared viewpoints with classical factor analysis. Its Bayesian factor analysis identifies them too and quantifies the uncertainty in each result.

3. **Native Transformation Project Intertribal Healthy Data Dashboard** 2025 - present
Open dashboard

An online platform for displaying community level health information and community level protective factors.

Packages

1. **bayesqm** Bayesian Q Methodology: Exact Rank-Order Likelihood for Forced Q Sorts 2026
CRAN · GitHub · Docs

An R package that models each forced Q sort exactly, as an ordered partition of the statements, and reports every Q-methodology result with its posterior uncertainty under one false-discovery rule. Its results are checked against planted truth in simulation and against the qmethod package where both compute the same result.

2. **actiRhythm** Circadian Rest-Activity Rhythm Analysis 2026
CRAN · GitHub · Docs

An R package that brings circadian methods usually spread across several packages into one interface: nonparametric measures, cosinor, periodograms, fractal and wavelet analysis, sleep scoring, and a hidden Markov rest-activity model. It reads ActiGraph, Axivity and GENEActiv files, and its results are checked against GGIR, cosinor, nparACT and ActCR.

3. **canhrcolors** CANHR Colors for R and Stata 2025
GitHub · Docs

An R package and Stata scheme that bring the Center for Alaska Native Health Research's 12-color palette into both programs: canhrcolors(), a ggplot2 color scale and a theme in R, and set scheme canhr in Stata, so figures from R and Stata share the same CANHR look.

4. **sleepwaker** Detect Sleep and Wake Periods from Accelerometry Data 2025
GitHub · Docs

An R package that inverts the sleep periods ActiGraph ActiLife detects into the waking periods between them, using the end of one sleep as the start of the wake and the start of the next sleep as its end. This keeps sleep out of sedentary time estimates.

TEACHING

University of Alaska Fairbanks Fairbanks, AK
STA 401: Regression and Analysis of Variance (TA + Lab Instructor) 2024 - 2025
Math 252: Calculus 2 (Teaching Assistant) 2023
STA 200X: Elementary Statistics (Teaching Assistant) 2023

University of Cape Coast Cape Coast, Ghana
STA 406: Time Series Analysis (Teaching Assistant) 2022
STA 301: Probability Distributions (Teaching Assistant) 2022
STA 101: Introduction to Probability (Teaching Assistant) 2022

PROFESSIONAL DEVELOPMENT

EPSCoR Respectful, Cooperative Research Workshop 04/2026
Alaska EPSCoR, University of Alaska Fairbanks Fairbanks, AK

Workshop on Environmental Research Priorities in Interior Alaska 02/2026
University of Alaska Fairbanks Fairbanks, AK

Summer Graduate School: Mathematics of Sea Ice and Polar Ecosystems 06/2025
Simons Laufer Mathematical Sciences Institute (SLMath) Fairbanks, AK

KU Probability and Statistics Conference 11/2024
University of Kansas Lawrence, KS

JSM Diversity Mentoring Program 08/2024
Joint Statistical Meetings, American Statistical Association Portland, OR

Multi-Species Occupancy Model Workshop 06/2024
Alaska Chapter, American Statistical Association Fairbanks, AK

LEADERSHIP AND SERVICE

University of Cape Coast Cape Coast, Ghana

Educational Head, Actuarial Club 2022

Launched and coordinated the Empowered Data Analyst Competition - a semester-end data analysis and visualization showcase that promoted statistical applications to local health, agriculture, and family welfare research.

Class Representative, Actuarial Science Students 2018 - 2022

Facilitated communication between students and faculty and disseminated departmental updates and decisions.

Co-Author, Departmental Solution Manual 2021 - 2023

Co-authored solution manuals for past examination questions in the department's actuarial science and statistics courses: ACT 212 (Mathematics of Investment and Credit), ACT 214 (Financial Analysis and Economics), ACT 301 (Life Contingencies I), ACT 304 (Life Contingencies II), and STA 306 (Multivariate Distributions).

TOOLS

Programming Languages	R, Stata, Python, C++, JavaScript, HTML, CSS
Statistical Software & Tools	RStudio, Git/GitHub, LaTeX, JAGS, Stan, KADE, SmartPLS 4, ActiLife, Mplus, MAXQDA, REDCap

MEMBERSHIPS

American Statistical Association (ASA) 2024 - present
Institute of Mathematical Statistics (IMS) 2024 - present
Society of Actuaries (SOA) 2022 - present